

Homework 3

1. Read “Eggs in Early Complementary Feeding and Child Growth: A Randomized Controlled Trial” by Iannotti et al. (2017), available on Brightspace.
 - a. Describe the study population. (1 point)
 - b. Describe the intervention and control treatments in this study. (2 points)
 - c. List the inclusion and exclusion criteria. (2 points)
 - d. Who among researchers or participants were blinded? (1 point)
 - e. List and describe two considerations the responsible IRBs may have considered when reviewing the protocol and tools for this study. (These may be considerations that would have been addressed on an informed consent form, for example.) (2 points)
 - f. The authors note, “Children in the egg group showed a higher prevalence of stunting and underweight than the control group at baseline.” Why was this surprising? (1 point)
 - g. How did the authors attempt to address the baseline imbalance in their analysis? (1 point)
 - h. Describe and discuss one limitation of this study. (1 point)
 - i. Describe one potential threat to the internal validity of this study. (2 points)
 - j. Describe one potential threat to the external validity of this study. (2 points)
 - k. What power and significance level were used for the power calculations? (1 point)
 - l. Find and review the entry for this trial at clinicaltrials.gov. What are the listed primary and secondary outcomes? How do these compare with the outcomes reported in the paper? (2 points)

NOTE: The following questions are not related to the Iannotti et al. paper.

2. What is one statistical risk of adding a new primary aim after a study has started? (3 points)
3. Your colleagues have conducted a randomized study, but they need help analyzing the data. They have asked you to join them as a collaborator. They will pay you for your time and include you as a co-author on their publications. You agree and start working on the data. The study participants were adults recently diagnosed with a chronic disease, and the study is evaluating a new treatment compared with the standard of care. As you get to know the study better, you learn that the participants were not actually randomized to the study arms; instead, the researchers, who are

physicians, examined each case and decided whether a participant should receive the new treatment or the standard of care. What do you do next? Explain your reasoning:

- a) Describe clearly what the ethical issue is here. Do you need any more information to understand the issue? If so, write a question you can ask your collaborators. (2 points)
 - b) What are some possible alternative actions you can take in this situation? (4 points)
 - c) Which alternative action would you choose? Which principles or frameworks would you use to make and justify your decision? (There is no single right answer here; you should explain your reasoning.) (4 points)
4. Select the vignette that you found most interesting from the two webinars assigned in Topic 3. Then answer the following:
- a) Who experienced the ethical challenge and which sector did they come from (academia/research, government, industry, ...)? (1 point)
 - b) Describe in your own words the ethical challenge they faced. (2 points)
 - c) How did they resolve the challenge? Which specific ethical guideline(s) informed how they handled the challenge, and how? (4 points)

Homework 3

Alexandra Chang

PUBH 526: Design and Analysis of Randomized Trials in Public Health

September 18, 2025

Read “Eggs in Early Complementary Feeding and Child Growth: A Randomized Controlled Trial” by Iannotti et al. (2017), available on Brightspace.

Question 1.a (1 point)

Describe the study population.

- Infants 6–9 months old and their mothers/primary caregivers from five rural parishes in Cotopaxi Province, Ecuador; $n = 163$ enrolled.
- Majority of households were rural, indigenous, or mestizo; most mothers were young (mean age ≈ 25) and infants were firstborns.
 - At baseline, $\approx 41\%$ were firstborn.

Question 1.b (2 points)

Describe the intervention and control treatments in this study.

- **Intervention:** One medium egg (≈ 50 g) daily for 6 months, provided weekly, with reminders and monitoring.
- **Control:** No eggs supplied; both groups received weekly home visits and social marketing for the “Lulun Project” to encourage participation.

Question 1.c (2 points)

List the inclusion and exclusion criteria.

- **Inclusion:** Participants are infants between 6 and 9 months old, singleton birth, in good health, and a caregiver willing to participate.
- **Exclusion:** Participants with congenital heart disease, acute severe malnutrition, and/or known egg allergy.

Question 1.d (1 point)

Who among researchers or participants were blinded?

- In this study, the field team investigators were masked to group assignments (except one staff member managing enrollment/logistics).
- Participants (caregivers and infants) could not be blinded due to the nature of the intervention (eggs are obvious).
- During data analysis, some investigators were blinded for sensitivity analyses.

Question 1.e (2 points)

List and describe two considerations the responsible IRBs may have considered when reviewing the protocol and tools for this study. (These may be considerations that would have been addressed on an informed consent form, for example.)

1. **Food safety/allergy risk:** Informed consent needed to cover monitoring for egg allergy or food intolerance.
2. **Caregiver burden & safety:** Weekly visits, blood draws, and child morbidity monitoring to ensure confidentiality, informed consent, and referral to care if illness arose.

Question 1.f (1 point)

The authors note, “Children in the egg group showed a higher prevalence of stunting and underweight than the control group at baseline.” Why was this surprising?

- At baseline, the egg group had higher stunting and underweight prevalence than controls. This may seem odd as randomization is expected to balance baseline characteristics across arms. Random imbalance in key outcomes undermines comparability.

Question 1.g (1 point)

How did the authors attempt to address the baseline imbalance in their analysis?

- To address the baseline imbalance in the analysis, the authors adjusted for baseline anthropometric measures, as well as age and sex of the child, using generalized linear models.

Question 1.h (1 point)

Describe and discuss one limitation of this study.

- Birth weight and gestational age were important determinants of growth; however, these factors were not measured in the study, which would limit the interpretation of growth trajectories.

Question 1.i (2 points)

Describe one potential threat to the internal validity of this study.

- A potential threat to the internal validity of this study would be the possible contamination/sharing, as eggs given to index child may have been shared with siblings, diluting the treatment effect.
- In addition, higher baseline diarrhea in the egg arm could confound results.

Question 1.j (2 points)

Describe one potential threat to the external validity of this study.

- A potential threat to the external validity of this study would be the limits regarding generalizability, as results apply to rural Ecuador with high stunting prevalence and cultural acceptance of eggs, but may differ in populations with low stunting, different dietary practices, or poor egg availability.

Question 1.k (1 point)

What power and significance level were used for the power calculations?

- Sample size calculations assumed an effect size of 0.35 SD, followed by an $\alpha = 0.05$, and power = 90% ($1 - \beta = 0.90$).

Question 1.l (2 points)

Find and review the entry for this trial at clinicaltrials.gov. What are the listed primary and secondary outcomes? How do these compare with the outcomes reported in the paper?

- ClinicalTrials.gov (NCT02446873):
 - **Primary outcomes:** Serum/plasma biomarkers such as vitamin B12, choline, betaine, polyunsaturated fatty acids, and amino acids.

- **Secondary outcomes:** Attitudes/beliefs about eggs, dietary diversity/frequency (including egg consumption), and anthropometry (LAZ, WAZ, WLZ, BMI-z).
- **Paper (Iannotti et al., Pediatrics 2017):**
 - Reported anthropometry (LAZ, WAZ, WLZ, BMI-z), stunting/underweight, dietary intake, and morbidity.
 - Biomarkers (the registered primary) were not in this article but appeared in a separate AJCN 2017 publication.
- **Comparison:**
 - The Pediatrics paper emphasized registered secondary outcomes (growth), while registered primary outcomes (biomarkers) were published separately—consistent with the paper’s statement that biomarker outcomes are reported in a separate publication and that the trial is registered under NCT02446873.

Question 2 (3 points)

What is one statistical risk of adding a new primary aim after a study has started?

- Adding a new primary aim after a study has may introduce the risk of inflating the Type I error rate, as it effectively increases the number of hypotheses being tested without appropriate statistical correction (e.g., for multiple comparisons).
 - Appropriate statistical corrections include adjusting α (e.g., Bonferroni, Scheffé), using group sequential designs, and applying alpha-spending functions to reduce the risk of false-positive results.
- Adding new primary aim(s) may also undermine the integrity of pre-specified analyses, as the original power calculations and sample size were not designed to detect effects for the new outcome.
 - This risks data dredging, also known as p-hacking, where researchers capitalize on chance findings through statistical manipulation of data to find patterns which can be presented as statistically significant, ultimately reducing the credibility and reproducibility of the study.

(**Homework 4**) Your colleagues have conducted a randomized study, but they need help analyzing the data. They have asked you to join them as a collaborator. They will pay you for your time and include you as a co-author on their publications. You agree and start working on the data. The study participants were adults recently diagnosed with a chronic disease, and the study is evaluating a new treatment compared with the standard of care. As you get to know the study better, you learn that the participants were not actually randomized to the study arms; instead, the researchers, who are physicians, examined each case and decided whether a participant should receive the new treatment or the standard of care. What do you do next? Explain your reasoning:

Question 3.a (2 points)

Describe clearly what the ethical issue is here. Do you need any more information to understand the issue? If so, write a question you can ask your collaborators.

- The ethical issue is that the study was being treated and presented as a randomized controlled trial (RCT), but participants were **not actually randomized**. Instead, physicians assigned treatment based on clinical judgment. This misrepresentation raised concerns about the scientific integrity of the study and may lead to invalid causal inferences.
- The study violated the **ASA's Ethical Guidelines** below:
 - **Principle A.4:** opposes selective interpretation of data or study design
 - **Principle B.1:** requires transparency about how data were generated
 - **Principle D.5:** protects the rights and expectations of research subjects, especially if informed consent was based on randomization.
- **Yes**, I would need more information to understand the issue. I would ask the question:
 - “Did the IRB protocol and informed consent documents mention randomization, and were participants being told that their treatment would be assigned by researchers who were physicians?”

Question 3.b (4 points)

What are some possible alternative actions you can take in this situation?

- **Clarify** from the study investigators to fully understand how treatment assignment occurred and how the study was described in the IRB protocol and consent forms.
- Recommend **reclassifying** the study as observational, since treatment was assigned by physicians rather than randomly. The analysis should then use methods appropriate for observational data, such as regression adjustment and/or propensity scores.

- Advise investigators to **inform** and update the IRB about parts that have been incorrectly represented as random.
- **Withdraw** from the project if the research team refuses to correct the misrepresentation, to avoid participating in unethical scientific conduct.

Question 3.c (4 points)

Which alternative action would you choose? Which principles or frameworks would you use to make and justify your decision? (There is no single right answer here; you should explain your reasoning.)

- I would start by having a direct conversation with the research team to confirm how treatment assignments were made and whether the study was ever described as randomized in the IRB or consent forms.
 - If the participants were not actually randomized, I would recommend changing the description of the study to an observational design and adjusting the analysis accordingly.
- If the team is open to making that correction, I would be willing to stay on as a collaborator. But if they insist on presenting the study as randomized when it wasn't, I would choose to step away from the project.
 - Misrepresenting the study could lead to misleading and unethical conclusions.
- My decision reflects the ASA principles and frameworks mentioned in *Topic 3*:
 - **Recognizing the issue, weighing options, minimizing harm, and acting with transparency.**
 - Being honest about how data were collected (**Principle B**), being accountable for the way research is presented (**Principle A**), and protecting the rights of participants (**Principle D**).

Select the vignette that you found most interesting from the two webinars assigned in *Topic 3*. Then answer the following:

The following answers are based on the first vignette, **Principles A, B, C, and D**.

Question 4.a (1 point)

Who experienced the ethical challenge and which sector did they come from (academia/research, government, industry, ...)?

- The ethical challenge was experienced by **Marsha Levinson**, a statistician who worked in the industry sector, specifically at Pfizer and later as a leader overseeing statistical teams.

Question 4.b (2 points)

Describe in your own words the ethical challenge they faced.

- Marsha described situations during the publication process for clinical trials, where statisticians were sometimes excluded from **authorship**, even when they had made major contributions to the study design, analysis, or interpretation.
- Instead, most were offered only in the acknowledgments section, or worse, nothing at all. Meanwhile, others with less direct contributions were included as co-authors, raising concerns about fairness, transparency, and professional recognition in collaborative research.

Question 4.c (4 points)

How did they resolve the challenge? Which specific ethical guideline(s) informed how they handled the challenge, and how?

- **Individual level:** Initially, Marsha handled these issues individually through conversations with collaborators to argue for the inclusion of statisticians as authors.
- **Organizational level:** Eventually, they took a more formal approach by adopting specific **authorship guidelines** based on established publication standards, such as ICMJE, and implementing an SOP that clarified the process for authorship decisions.
 - Marsha concluded by saying that having procedures helped ensure that statisticians were **appropriately credited** for their work across projects.
- Marsha's decision reflects the ASA principles and frameworks mentioned in *Topic 3*:
 - **Principle A:** Accurately represent the team's contributions and resist unfair authorship exclusion.

- **Principle B:** Proper authorship to achieve transparency on responsibility for the statistical methods and analyses in the publication.
- **Principle C:** SOP ensured fair recognition and respect for all members involved for clear communication with external collaborators.
- **Principle D:** Recognizing authors promotes research accountability to protect the integrity of research findings shared with the public and research participants.

Homework 3 Solutions

1. Read “Eggs in Early Complementary Feeding and Child Growth: A Randomized Controlled Trial” by Iannotti et al. (2017), available on Brightspace.

- a. Describe the study population. (1 point)

Infants (6-9 months) living in rural Cotopaxi Province, Ecuador.

In general, it's good to include key demographic characteristics when describing the population (sex, age, etc., as mentioned in the inclusion/exclusion criteria). In this case, for example, the children's age range is relevant, as a similar intervention for older children may not have the same effect.

It would be useful to know how the 5 parishes were chosen – randomly or based on certain criteria? – as that can affect the generalizability of the findings. Similarly, it would be useful to know if the staff found all mother-child pairs in the study area and screened them for eligibility, or if some selection happened there as well.

- b. Describe the intervention and control treatments in this study. (2 points)

The intervention consisted of one medium-sized egg per day for six months, delivered on a weekly basis with a reminder to mothers/caregivers to feed 1 egg/day to the index child. [Note that in describing an intervention, the duration of that intervention is important to mention. For public health interventions, it's also important to think about delivery method and messaging, as well as the main intended intervention.]

The control treatment involved no intervention. Families were visited on a weekly basis over the six months, but they did not receive any particular messages.

Children in both arms received referrals in the case of serious illness. It's not clear whether the social marketing strategy was part of the intervention, or whether that strategy also reached control group members or other community members that were not study participants.

- c. List the inclusion and exclusion criteria. (2 points)

Inclusion criteria: infant aged 6 to 9 months at enrollment, singleton birth, and in good health

Exclusion criteria: infant has a congenital heart condition, severe acute malnutrition, or an egg allergy

Studies will often also list informed consent as a criterion and specify whether that consent is verbal or written. Parent(s) provide informed consent for infants and young children (whether one parent is enough or both parents are needed depends on the potential risks and benefits of participating in the study).

- d. Who among researchers or participants were blinded? (1 point)

The paper states that investigators in the field were blinded, except for one person responsible for enrolling and monitoring participants. It is not clear who delivered the intervention, as they would clearly not be blinded. Participants could not be blinded (they would know whether they received eggs or not, but it's possible that baseline data were collected before participants were informed of their group assignment). There is also mention of blinding during sensitivity analyses, but that is not discussed later in the paper. The phrase "masked to group assignment" may mean that that an individual is completely blinded or knows coded group assignments.

- e. List and describe two considerations the responsible IRBs may have considered when reviewing the protocol and tools for this study. (These may be considerations that would have been addressed on an informed consent form, for example.) (2 points)

Several answers are possible here: the IRB would review issues related to potential risks, potential benefits, whether the proposed sample size would provide adequate statistical power to investigate the primary aims, whether measures are in place to protect participants' privacy, etc.

- f. The authors note, "Children in the egg group showed a higher prevalence of stunting and underweight than the control group at baseline." Why was this surprising? (1 point)

With randomization, we expect the two groups to be similar on average (i.e., we expect to have similar rates of stunting and underweight). The description of the randomization procedure does not suggest any possible causes of baseline imbalance. (There is always a small chance that some aspect of the randomization that was not described led to the imbalance.) The imbalance may be due to chance (which is more possible with small sample sizes). Very few randomized children were excluded from the baseline analysis, so missing data or dropouts would not explain the imbalance at baseline.

- g. How did the authors attempt to address the baseline imbalance in their analysis? (1 point)

For each outcome, in addition to the intention to treat analysis, they also estimated adjusted regression models that controlled for child age, sex, and baseline measure. (This can be helpful, but may not necessarily solve the problem.)

- h. Describe and discuss one limitation of this study. (1 point)

Several answers are possible here: the baseline imbalances for key outcomes; the study was originally powered for different outcomes than were reported in this paper; consumption of eggs was based on caregivers' reports and actual consumption could not be measured;

- i. Describe one potential threat to the internal validity of this study. (2 points)

Several answers are possible here: the two arms may have been imbalanced with respect to other (unmeasured) factors related to anthropometry (e.g., households with stunted children may also be more food insecure, affecting the response of those households to the intervention); caregivers may not feed all eggs to the index child as directed; parents in the intervention group may change other parts of their child's diet since they know their child is at least getting eggs.... Lack of blinding could also lead to changes in behaviors or practices that could ultimately threaten internal validity.

- j. Describe one potential threat to the external validity of this study. (2 points)

Several answers are possible here: other target populations may vary in degrees of malnutrition – perhaps a less malnourished population will see smaller benefits from increased egg consumption; other target populations may have different cultural views about the appropriateness/acceptability of eggs for very young children;

- k. What power and significance level were used for the power calculations? (1 point)

0.05 significance level and 90% power (applied to outcomes that aren't the subject of this paper)

- l. Find and review the entry for this trial at clinicaltrials.gov. What are the listed primary and secondary outcomes? How do these compare with the outcomes reported in the paper? (2 points)

<https://clinicaltrials.gov/ct2/show/study/NCT02446873>

Primary outcomes:

Serum/plasma concentrations of B12, choline, and betaine

Serum/plasma concentrations of polyunsaturated fatty acids (PUFAs) and amino acids

Secondary outcomes:

Attitudes, beliefs, and use of eggs in Pastocalle during the complementary feeding period

Frequency of consumption of eggs, other animal source foods, and dietary diversity in the child's diet in Pastocalle

The listed primary outcomes are not reported in this paper. Changes in anthropometry may not have been anticipated / hypothesized prior to the start of the study.

NOTE: The following questions are not related to the Iannotti et al. paper.

2. What is one statistical risk of adding a new primary aim after a study has started? (3 points)

Different answers are possible here:

The study should be adequately powered to answer all primary aims. The new aim may require a larger sample size than the study team had planned.

Also, new aims and research questions posed after a study starts are often driven by the study experience or a look at data collected so far. You may (consciously or unconsciously) propose a new aim that leans in the direction that the study seems to be going, and this can increase your false positive rate. For example: you're doing a study on food insecurity, and by the middle of the study, you're seeing changes in children's body weight. You start getting curious about that, and you propose an aim to look at the effects of your intervention on child body weight. However, because you're being driven by your experience and data so far (which suggest that something is happening with body weight), your chance of a false positive for this new aim increases.

Even without the above, the more aims you have (the more statistical tests you want to do), the greater the chance of at least one type I error.

3. [This question was moved to Homework 4.]
4. Select the vignette that you found most interesting from the two webinars assigned in Topic 3. Then answer the following:

All your answers will be different here!

- a) Who experienced the ethical challenge and which sector did they come from (academia/research, government, industry, ...)? (1 point)
- b) Describe in your own words the ethical challenge they faced. (2 points)
- c) How did they resolve the challenge? Which specific ethical guideline(s) informed how they handled the challenge, and how? (4 points)